

PATENT A

Self-Sustaining Neural-Motor Energy Harvesting Loop for Autonomous Cognitive Systems

Drawing Sheets: 8 figures

FIG. 1 — System Architecture Block Diagram (8-Layer Loop)

FIG. 2 — Energy Balance Comparison

FIG. 3 — SNN Architecture (LIF Neuron Model)

FIG. 4 — Energy Harvesting Circuit

FIG. 5 — Activity-Dependent Energy Dynamics

FIG. 6 — Hardware Reference Design

FIG. 7 — Validation Results Summary

FIG. 8 — Energy-Bounded Recursive Control Architecture

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

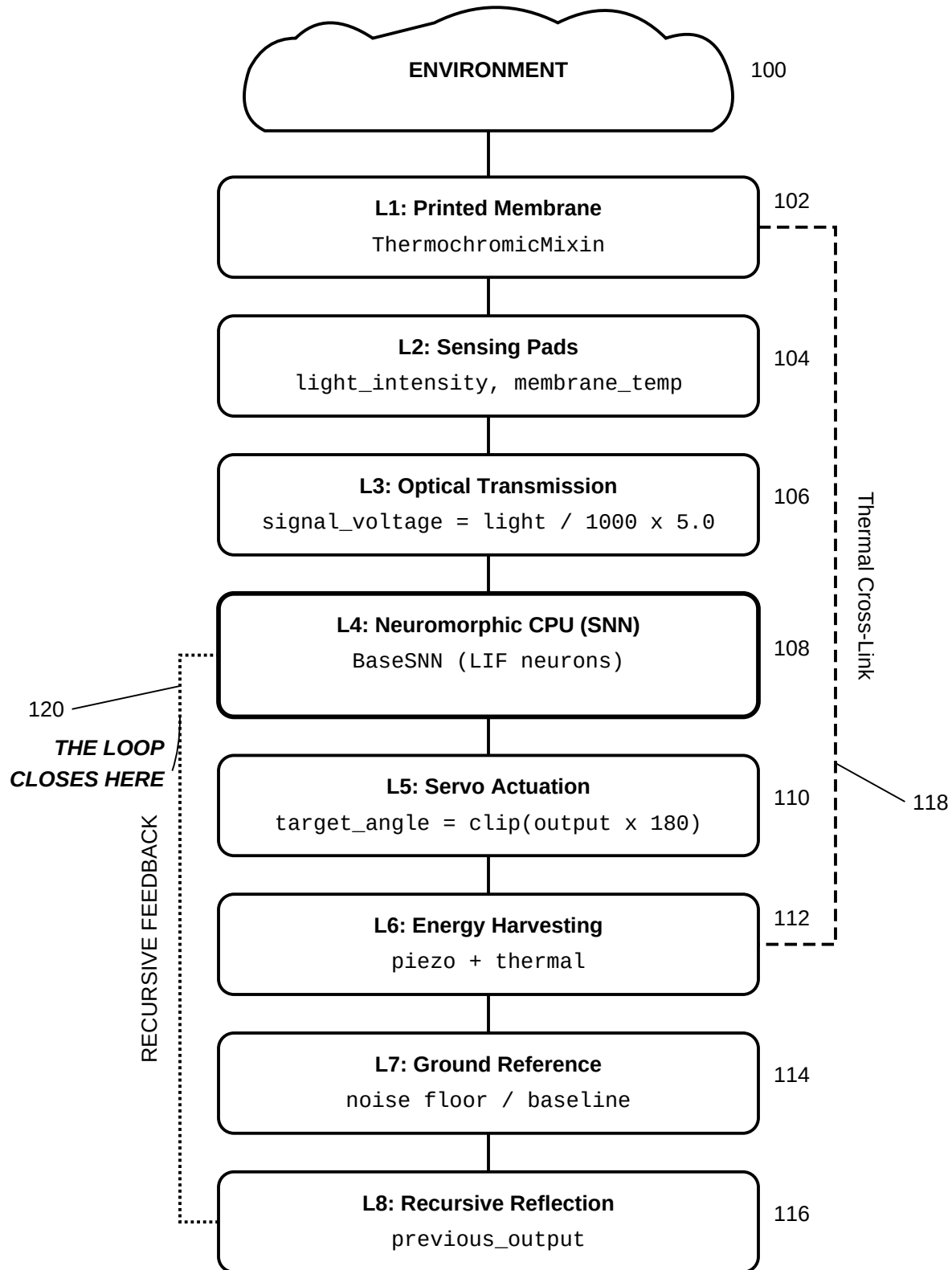


FIG. 1

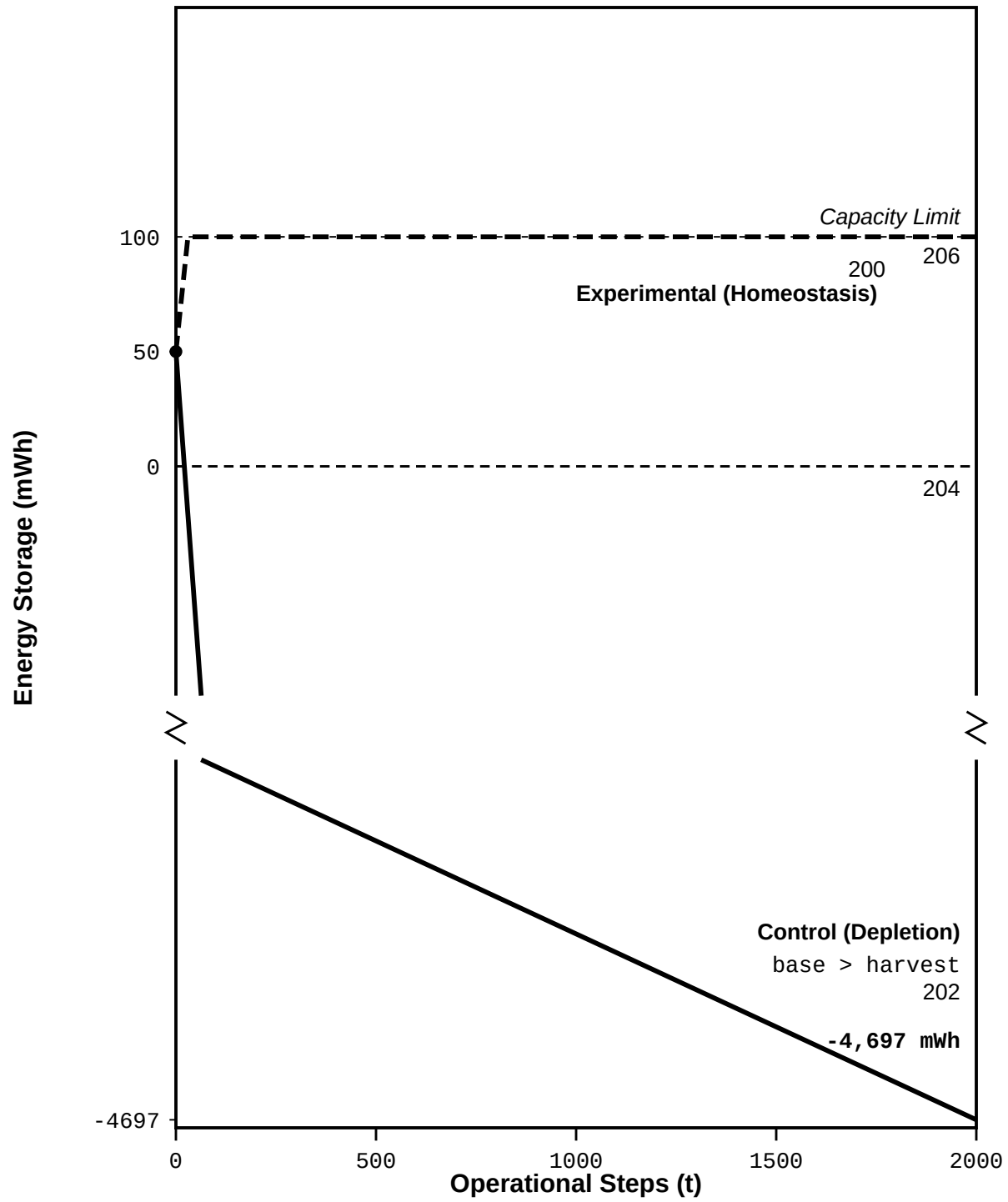
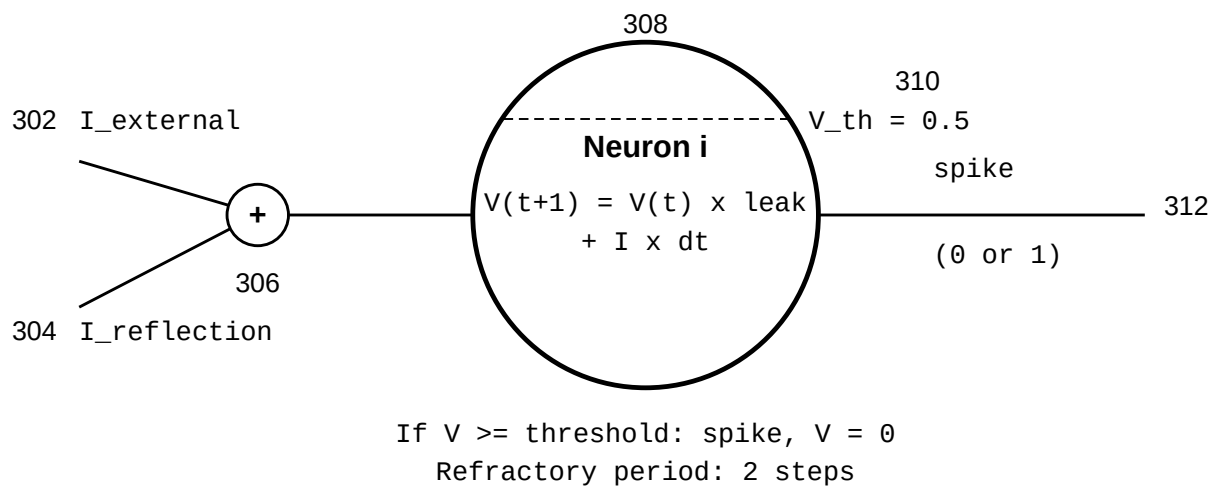
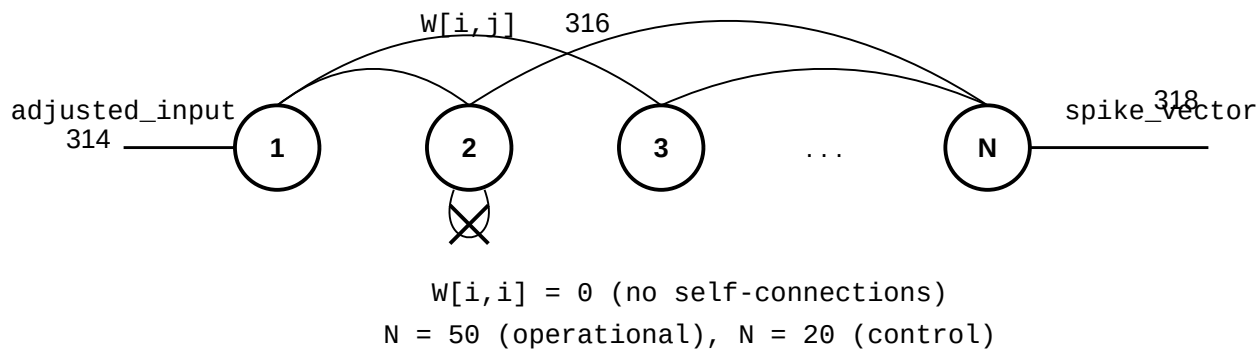


FIG. 2

Single LIF Neuron Detail



Network Topology

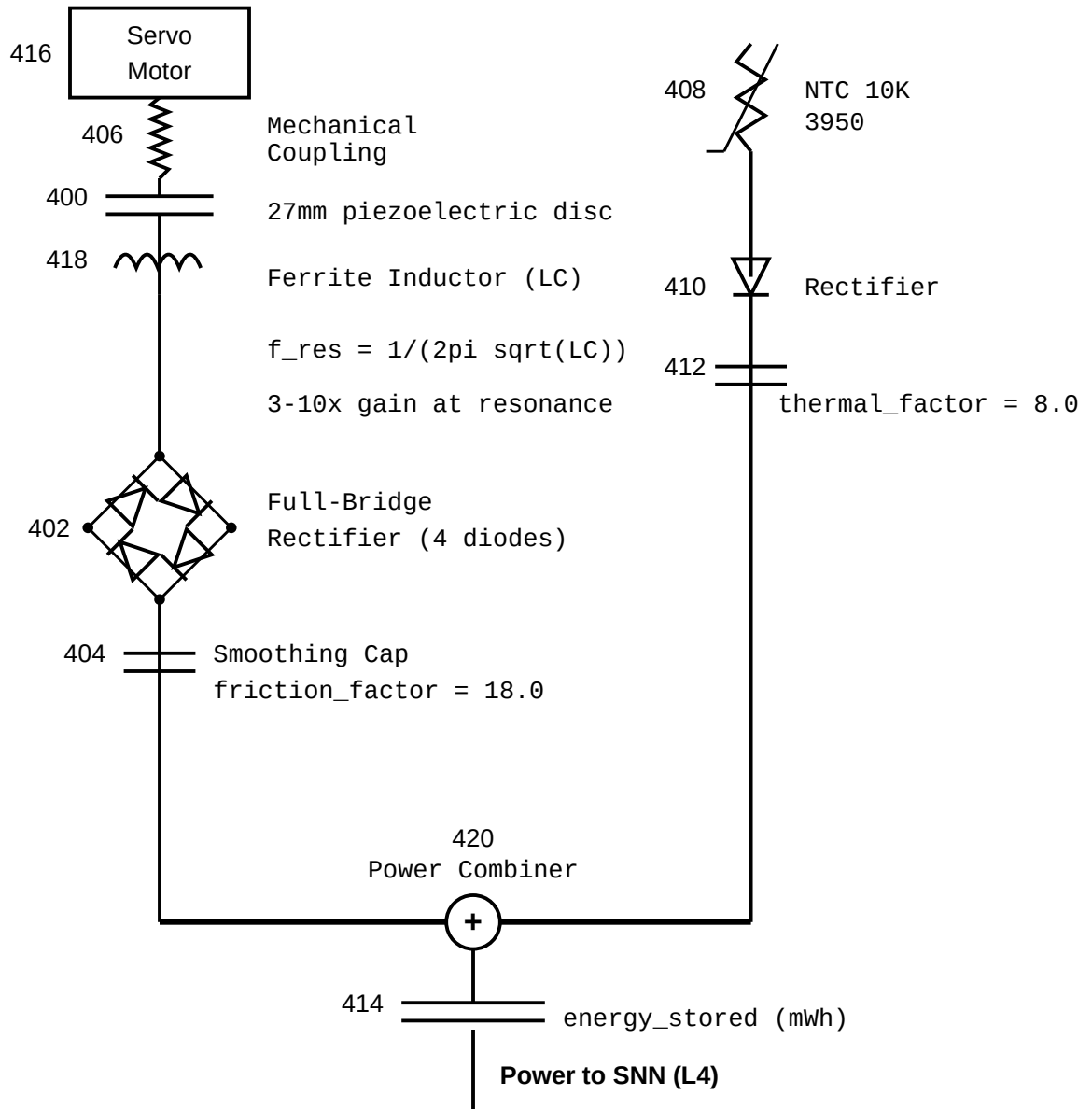


STDP Learning Rule

Post fires: $W += A+ \times \text{pre_trace}$ (LTP)
Pre fires: $W -= A- \times \text{post_trace}$ (LTD)
 $A- > A+$ (stability bias)

320

FIG. 3

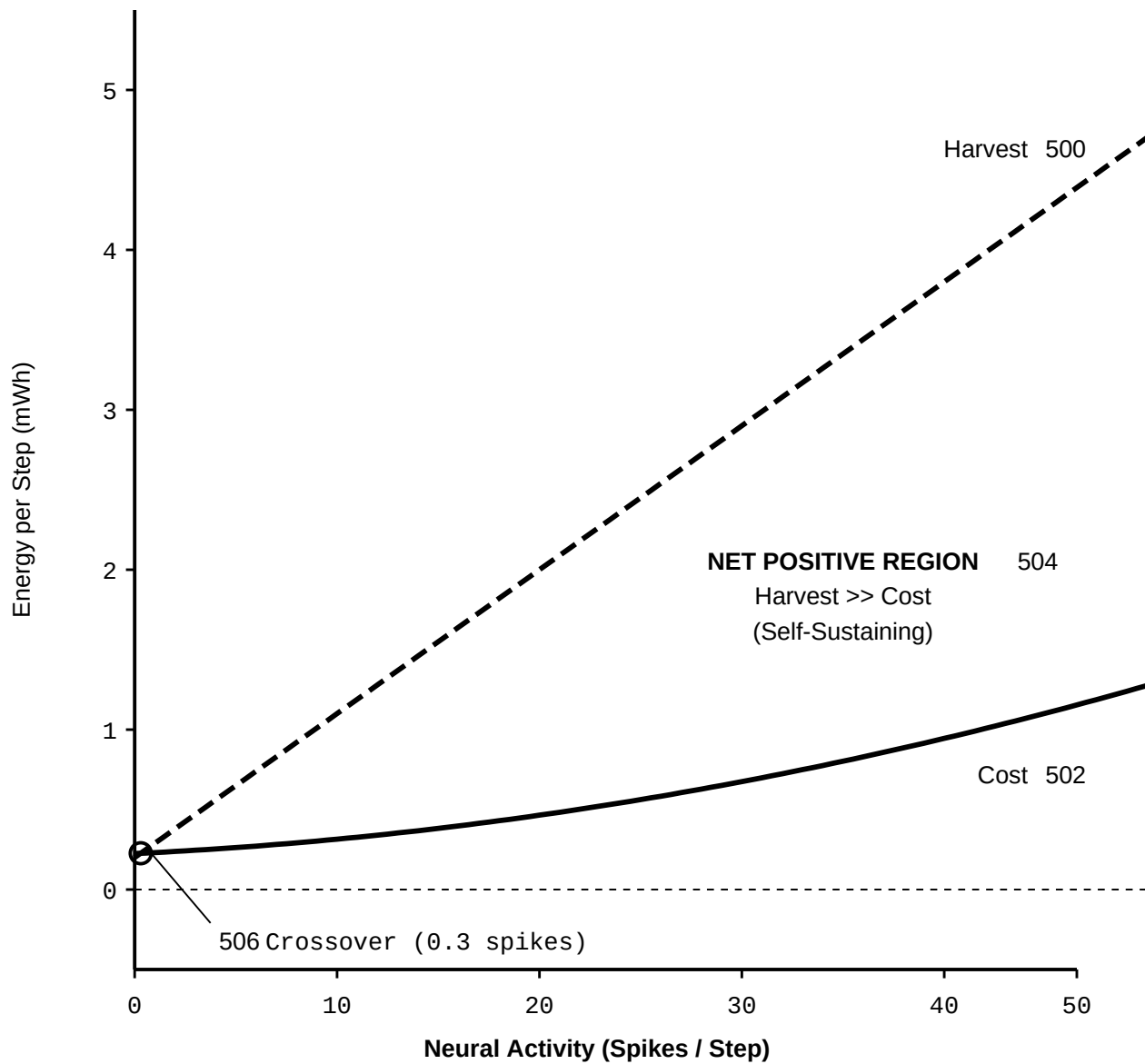
Piezoelectric Harvester**Thermoelectric Harvester****Power Budget**

Harvest = friction_harvest + thermal_harvest

Cost = base_consumption + activity_cost x spike_count²

Net = Harvest - Cost

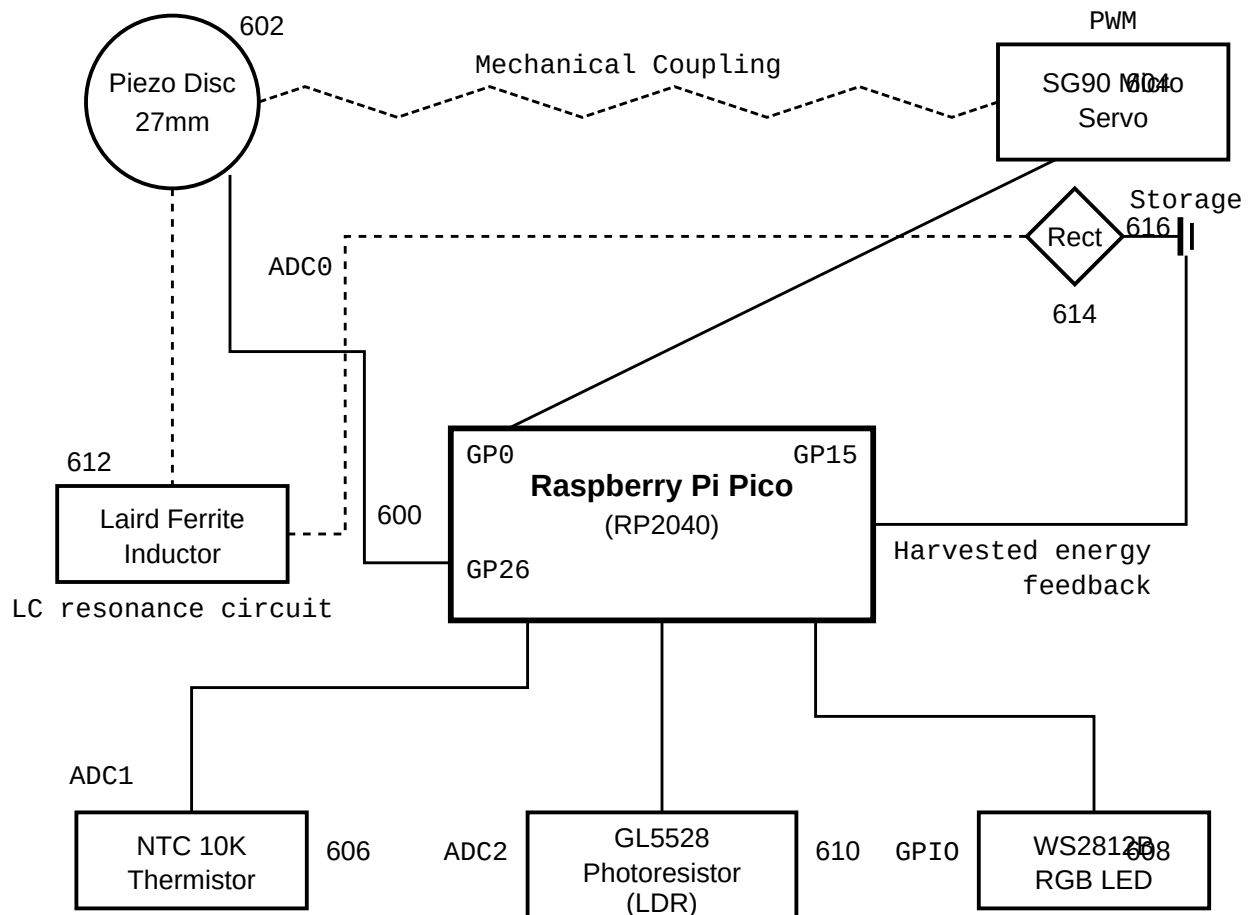
FIG. 4



$$\begin{aligned} \text{Harvest} &= 0.20 + (0.09 \times \text{spikes}) \text{ mWh/step} \\ \text{Cost} &= 0.225 + (0.006 \times s) + (0.0003 \times s^2) \text{ mWh/step} \\ \text{Net} &= \text{Harvest} - \text{Cost} \end{aligned}$$

Zone 1: Quiescent Drain
0 - 1 Spikes: Cost > Harvest

FIG. 5



Component Summary

- Raspberry Pi Pico (RP2040)
- SG90 Micro Servo
- 27mm Piezoelectric Disc
- NTC 10K Thermistor
- Photoresistor (LDR)
- WS2812B RGB LED
- Misc (wires, board, inductor)

—— Signal / Control

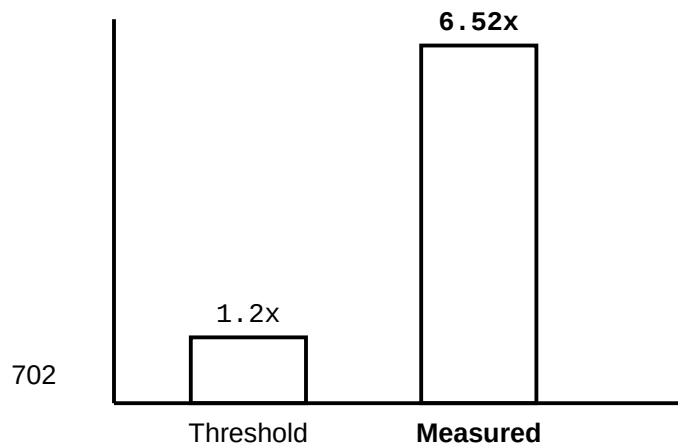
----- Energy Harvesting

FIG. 6

700

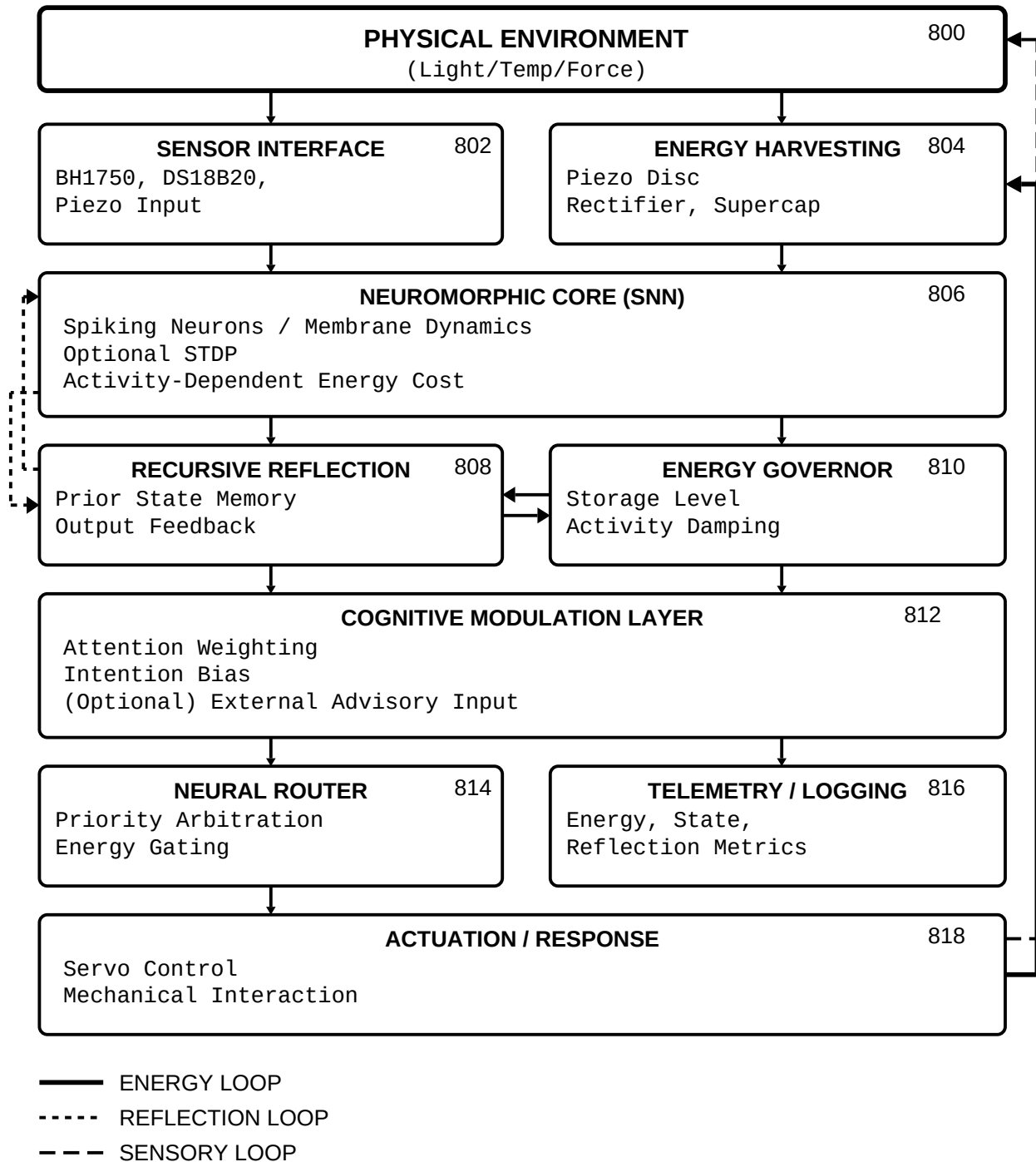
VALIDATION RESULTS SUMMARY

Claim	Criterion	Measured	Threshold	Result
Phase 7 Control Baseline (EnergyConfig)				
A	Closed-loop continuity	2000 pts	2000	PASS
B	Boundedness	bounded	bounded	PASS
C	Robustness (noise)	Std(theta) <= 45	<= 45	PASS
D	Input-output gain	corr >= 0.2	>= 0.2	PASS
E	Saturation ratio	sat <= 0.20	<= 0.20	PASS
Phase 7 Full Integration (Control confirms energy depletion)				
--	Energy at 2000 steps	-4,697 mWh	< 0	CONFIRMED
MCP Integration (BalancedEnergyConfig)				
--	Energy at 2000 steps	100.0 mWh (capped)	> 0	CONFIRMED
Phase 8 STDP				
F8.1	Weight entropy decrease	2.95 to 2.02 bits	decrease	PASS
F8.2	STDP MI > 1.2x frozen	6.52x	> 1.20	PASS
F8.3	Weight convergence	0.00%	< 10%	PASS

STDP Learning Efficacy (Claim F8.2)

All claims validated. System is reproducible via git tags.

FIG. 7



All computation is constrained by real-time energy availability derived from mechanical interaction.

FIG. 8

PATENT B

Configurable Recursive Self-Observation Method and System for Spiking Neural Networks with Dynamic Reflection Coefficient Modulation

Drawing Sheets: 6 figures

FIG. 1 — Self-Observation Feedback Loop

FIG. 2 — Reflection Coefficient Spectrum

FIG. 3 — Dynamic Modulation Sources

FIG. 4 — Self-Referential Learning Loop (STDP + Self-Observation)

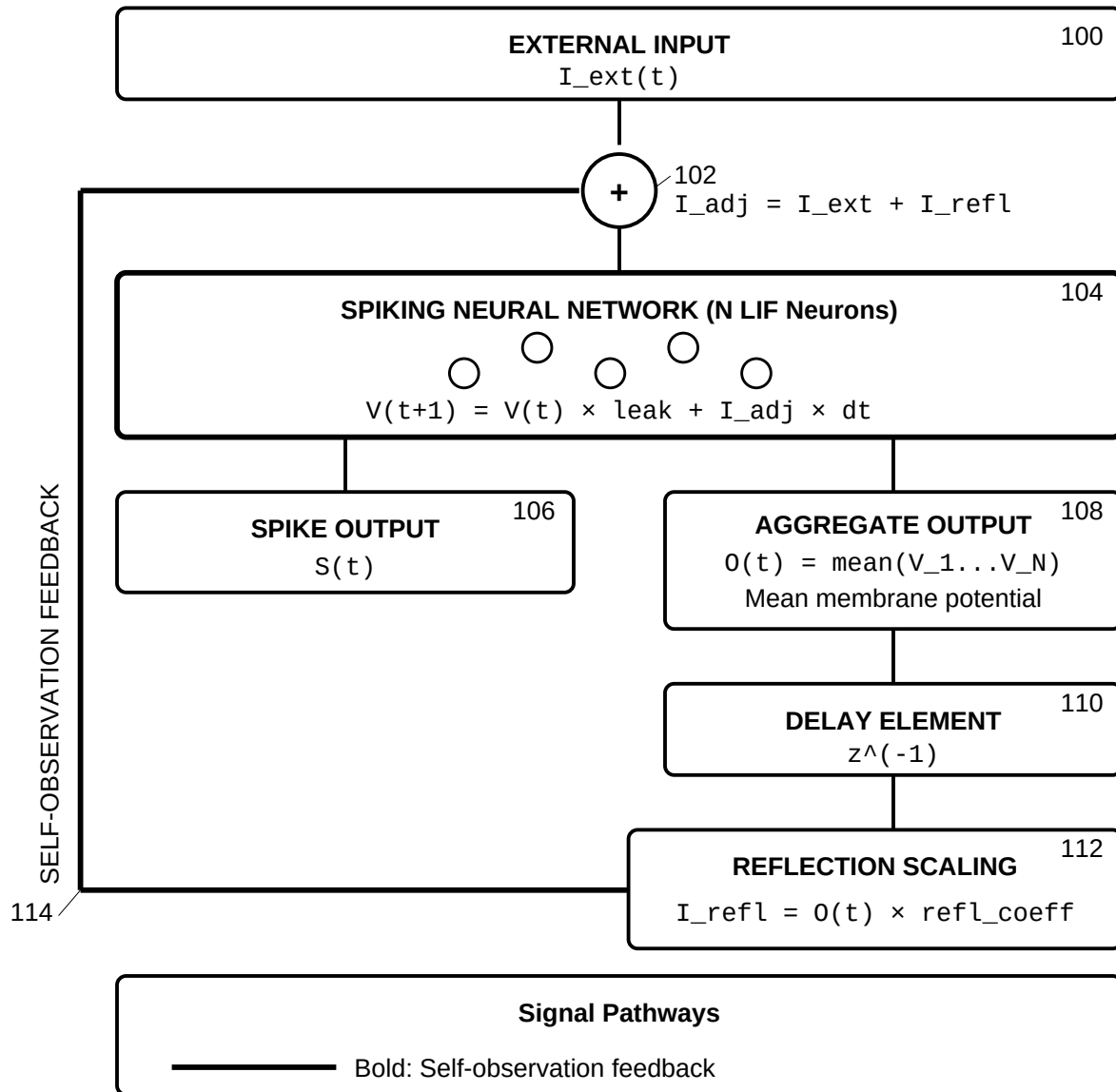
FIG. 5 — Energy-Aware Self-Observation Regulation

FIG. 6 — End-to-End Signal Flow with Self-Observation Integration

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

SELF-OBSERVATION FEEDBACK LOOP
Core Mechanism for Recursive Self-Observation

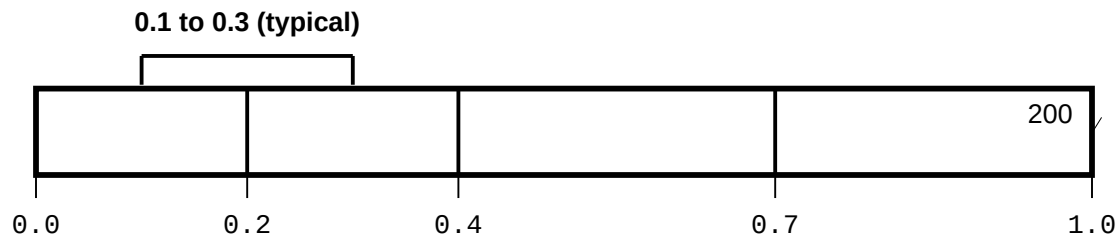


Self-observation: network observes its own prior state.
Reflection coefficient: 0.0-1.0. Operational: 0.1-0.3.

FIG. 1

REFLECTION COEFFICIENT SPECTRUM

Behavioral Effects Across Full Range (0.0 to 1.0)



Region 1 (0.0): No Self-Observation

202 ✓

Pure feedforward processing

Behavior: System responds only to external input

Region 2 (0.0 - 0.2): Minimal Self-Awareness

204 ✓

Slight influence of prior state

Behavior: Primarily externally driven

Region 3 (0.2 - 0.4): Balanced Self-Observation

206 ✓
OPERATIONAL RANGE

External input + meaningful self-reference

Default: $\text{refl_coeff} = 0.2 + \text{modulation} \times 0.1$

Region 4 (0.4 - 0.7): Self-Dominated Processing

208 ✓

Internal state dominates

Behavior: Risk of echo chamber dynamics

Region 5 (0.7 - 1.0): Self-Absorption

210 ✓

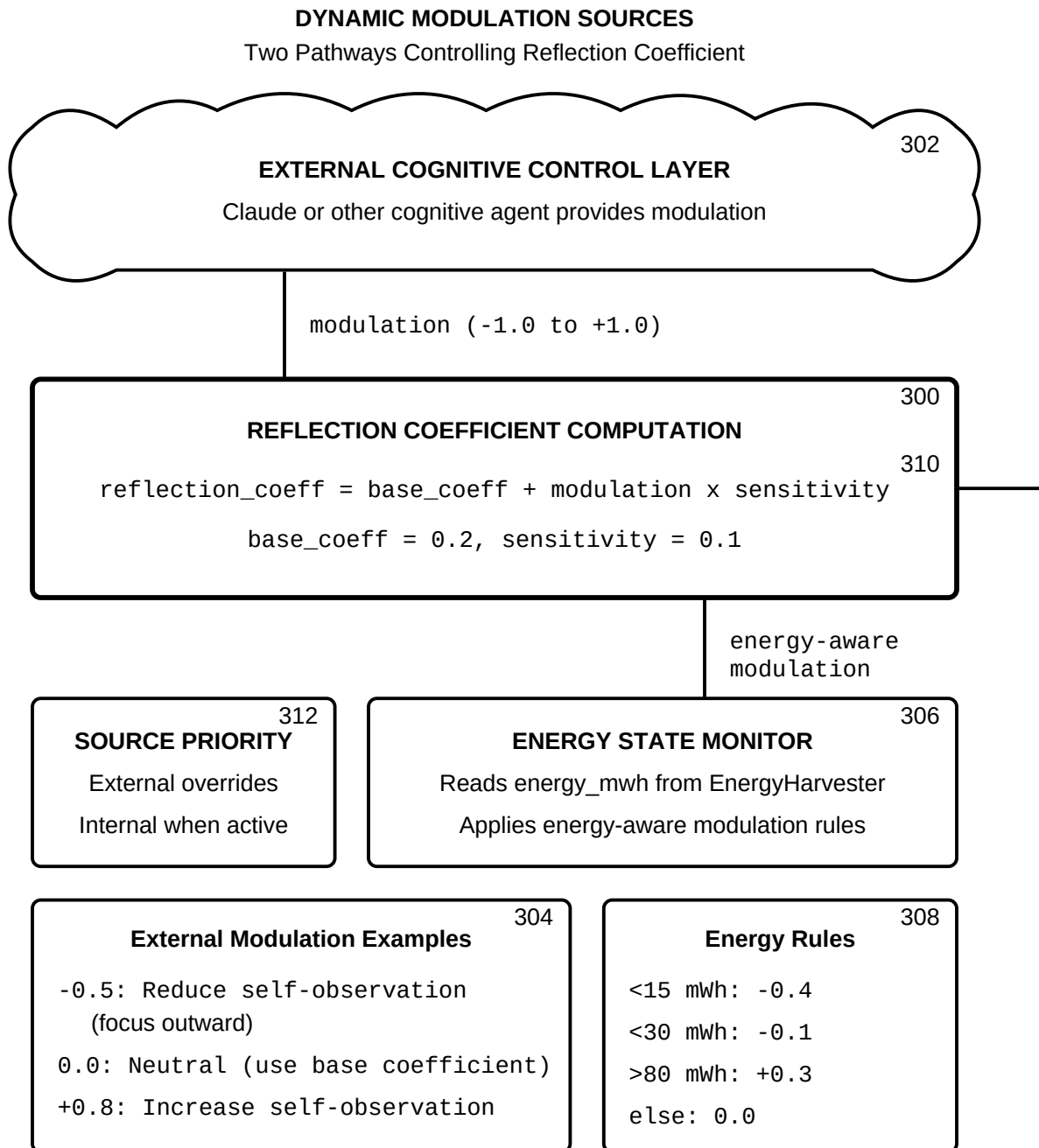
External input overwhelmed

Behavior: System locks onto its own prior state

Increasing self-referential processing

The reflection coefficient is the system's self-awareness dial

FIG. 2

**FIG. 3**

SELF-REFERENTIAL LEARNING LOOP
(STDP + Self-Observation)

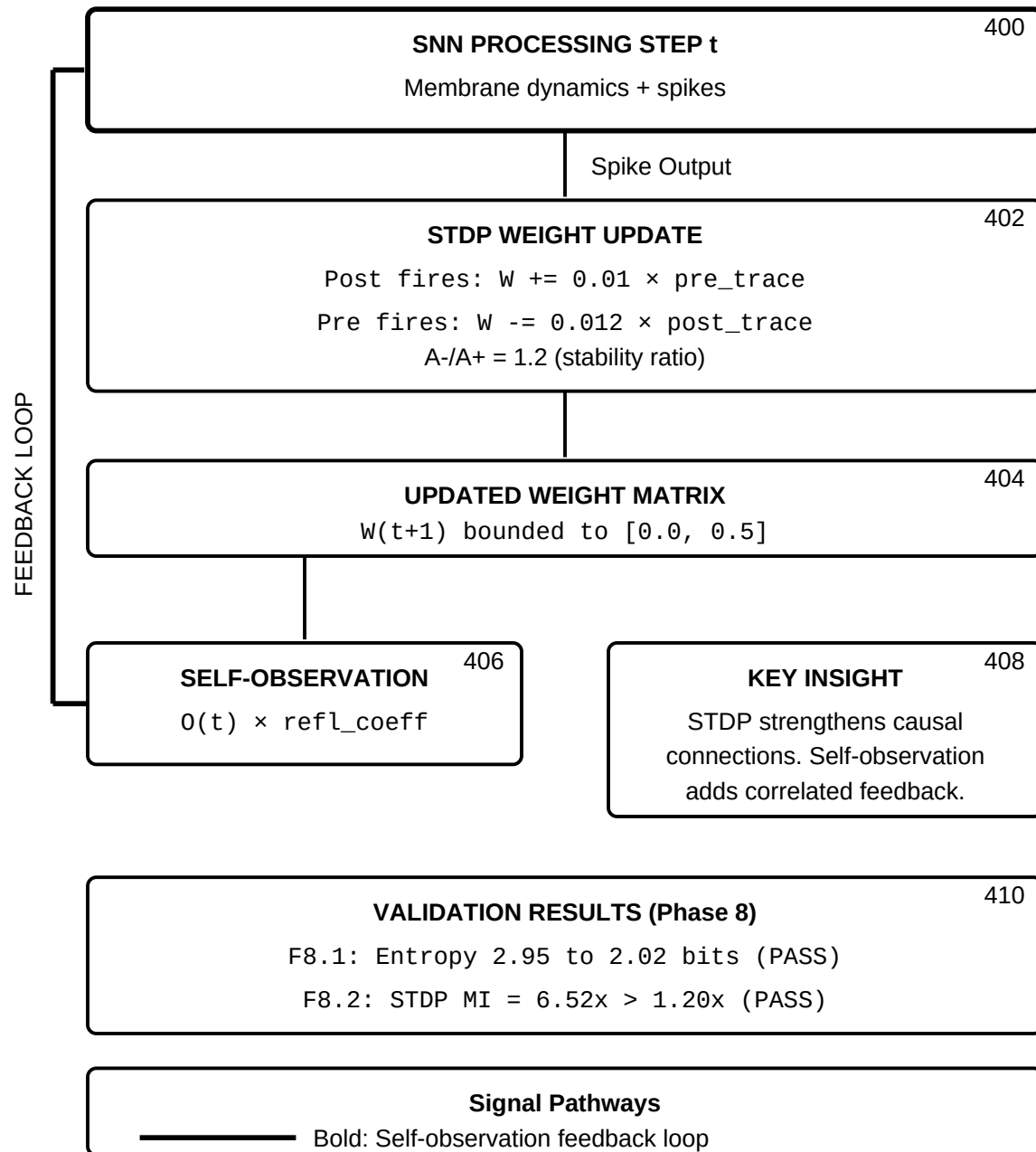
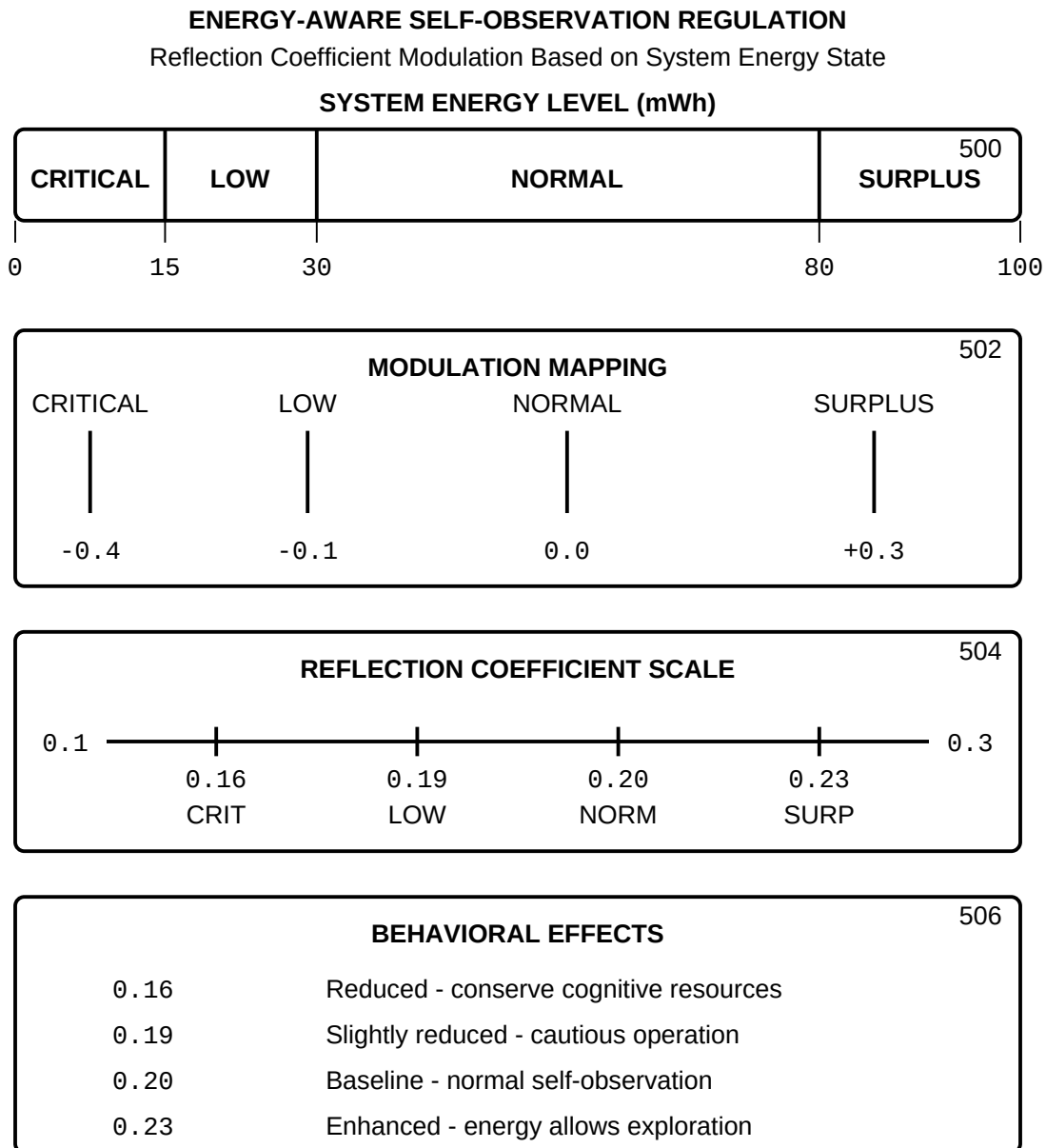


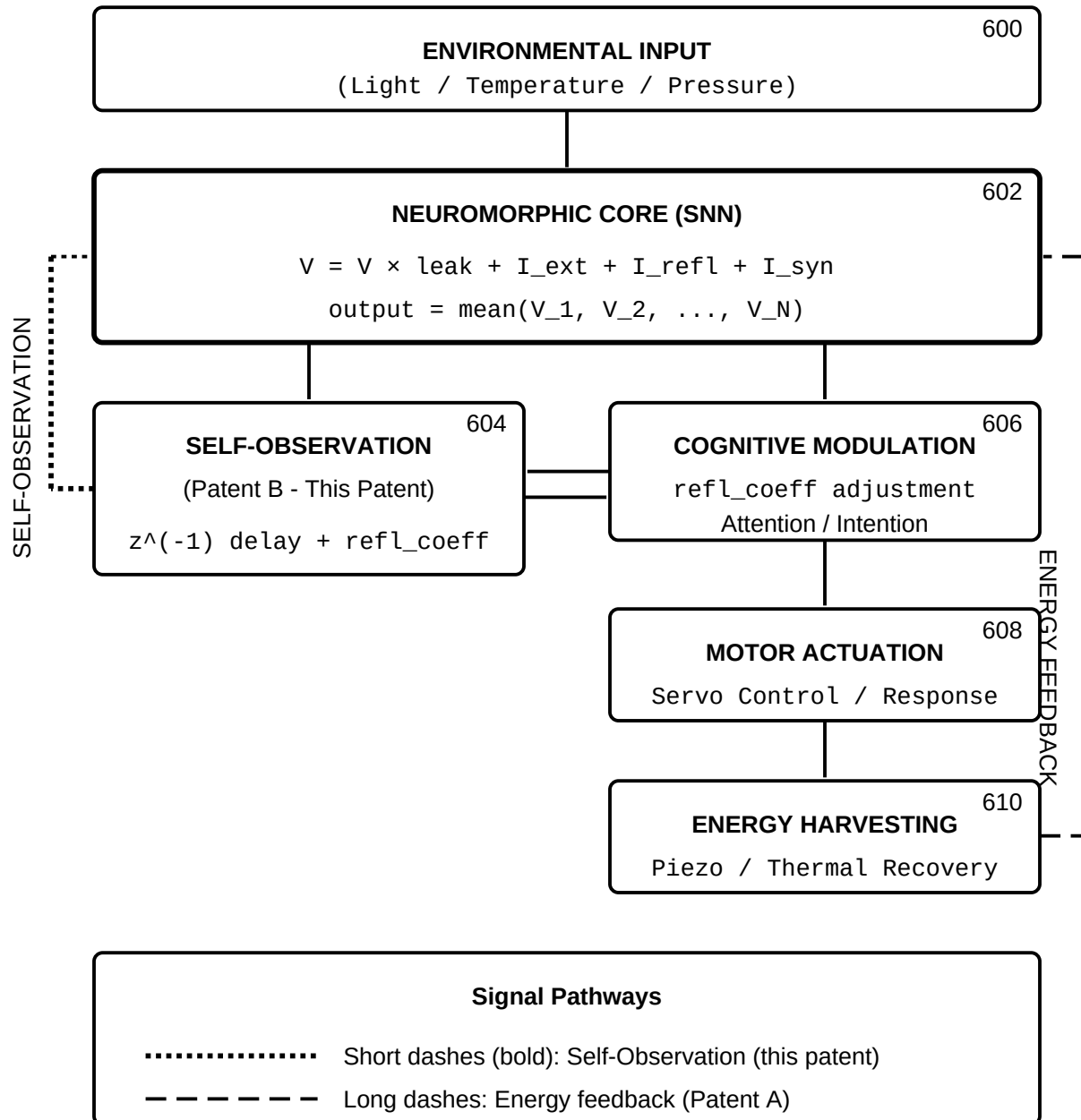
FIG. 4



Energy-aware regulation ensures efficient cognitive resource management.
Low energy reduces self-observation; surplus energy enables exploration.

FIG. 5

END-TO-END SIGNAL FLOW
Self-Observation Integration in Complete System



The self-observation pathway feeds the network's own aggregate state back as explicit input at configurable gain (0.0 to 1.0).

FIG. 6

PATENT C

Method and System for Autonomous Self-Regulation and Cognitive Resynchronization in Neural Processing Systems During Disconnection from External Control Layers

Drawing Sheets: 7 figures

FIG. 1 — System Architecture with Fallback

FIG. 2 — State Transition Diagram

FIG. 3 — Energy-Aware Modulation Curve

FIG. 4 — Autonomous Input Generator Output

FIG. 5 — Resynchronization Payload Structure

FIG. 6 — Recovery Timeline Diagrams

FIG. 7 — End-to-End Signal Flow: Connected vs. Autonomous

Inventor: Kevin Christopher Ward

Provisional Patent Application — Filed February 2, 2026

SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK

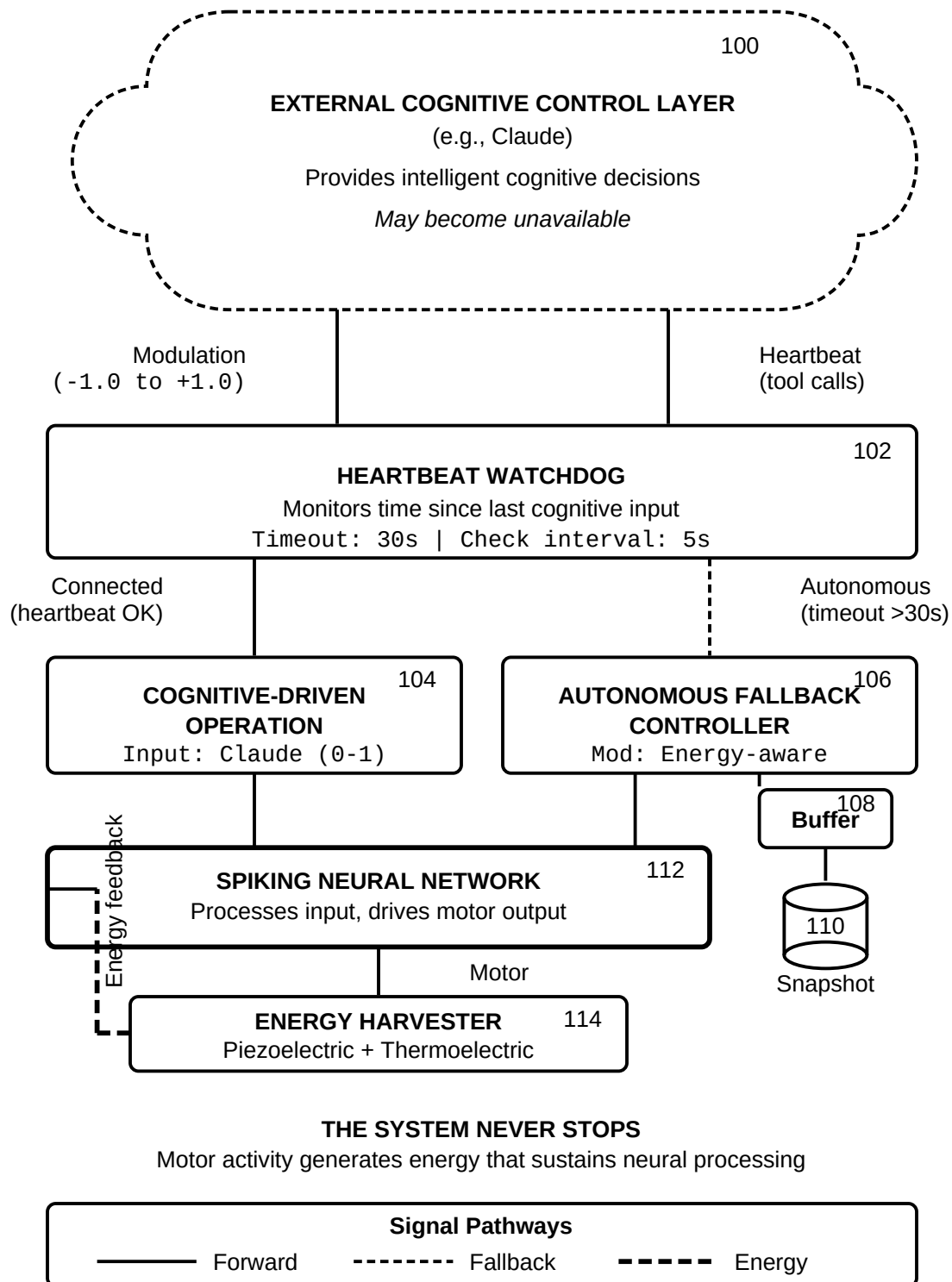
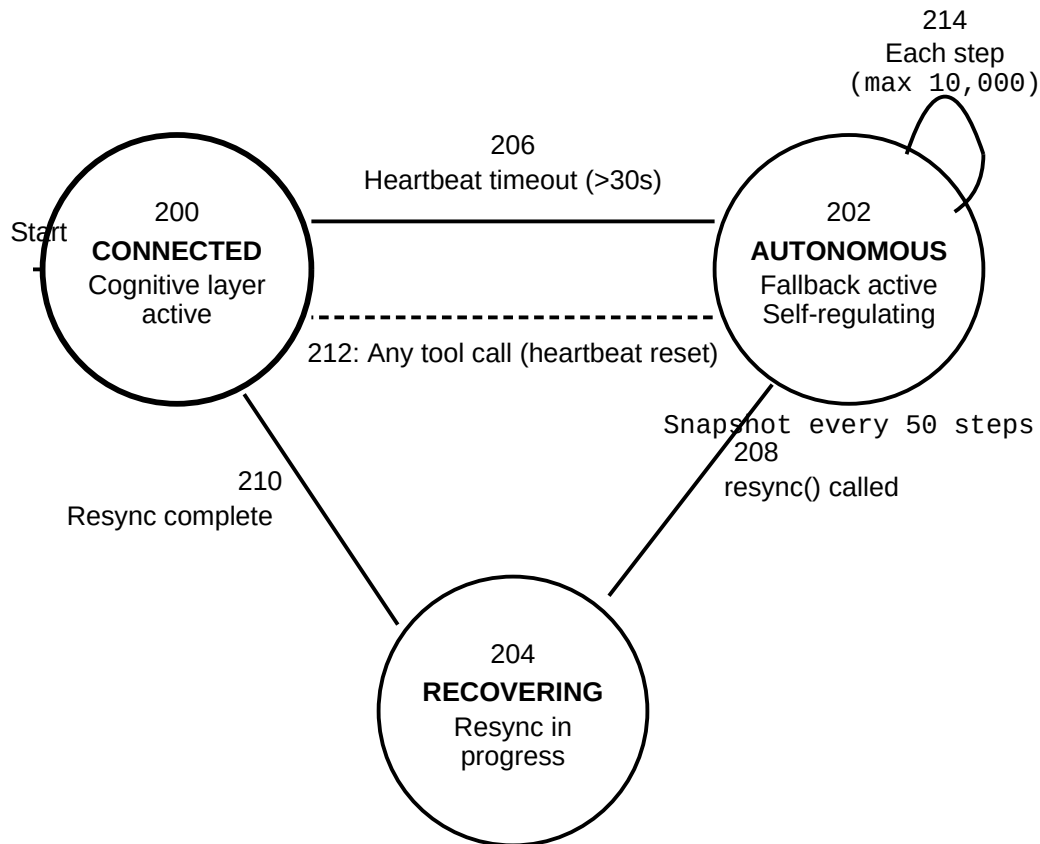


FIG. 1

STATE TRANSITION DIAGRAM



Transition Summary

206: Timeout triggers fallback

212: Direct reconnection

208: Explicit resync request

214: Autonomous step loop

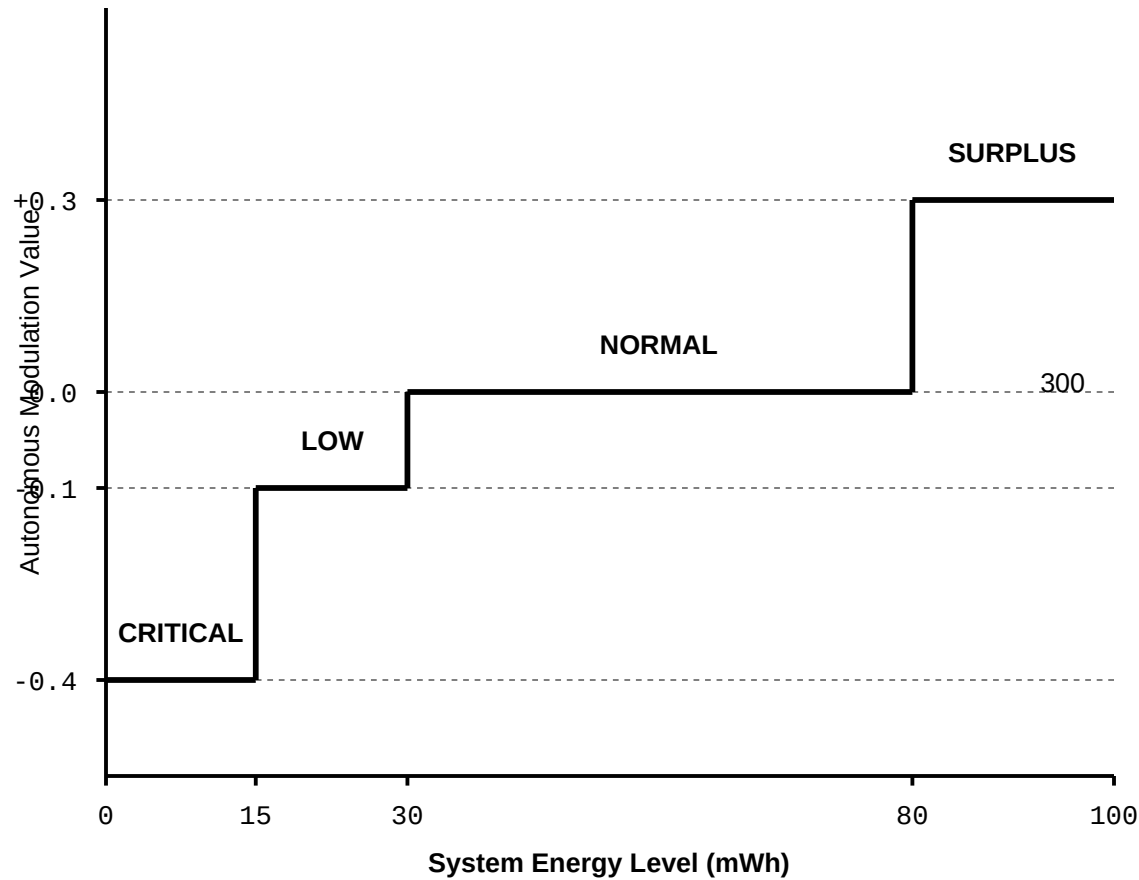
210: Resync completes

— Initial state

- - - - - Direct reconnect

FIG. 2

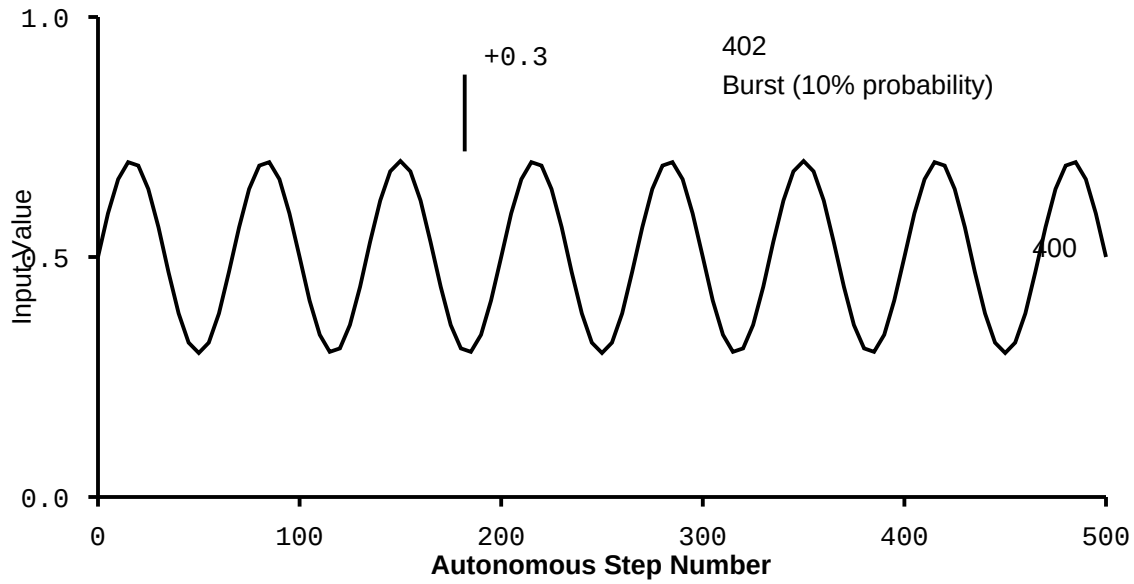
ENERGY-AWARE MODULATION CURVE



Behavioral Effects			302
< 15 mWh	mod = -0.4	Maximum conservation	
15-30 mWh	mod = -0.1	Cautious operation	
30-80 mWh	mod = 0.0	Standard processing	
> 80 mWh	mod = +0.3	Exploration enabled	

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



404

Autonomous Input Formula

$$\text{circadian} = 0.5 + 0.2 \times \sin(2\pi \times t \times 3)$$

burst = 0.3 (with 10% probability per step)

Energy Recovery During Autonomous Operation

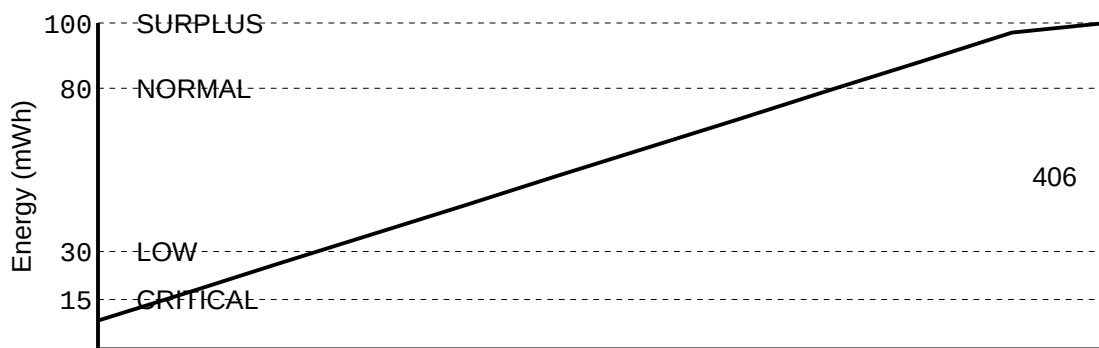


FIG. 4

RESYNCHRONIZATION PAYLOAD STRUCTURE

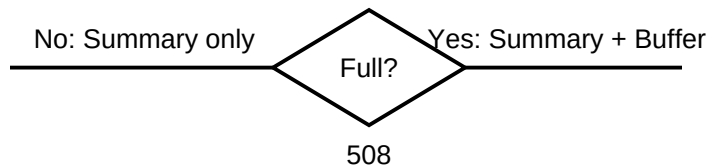
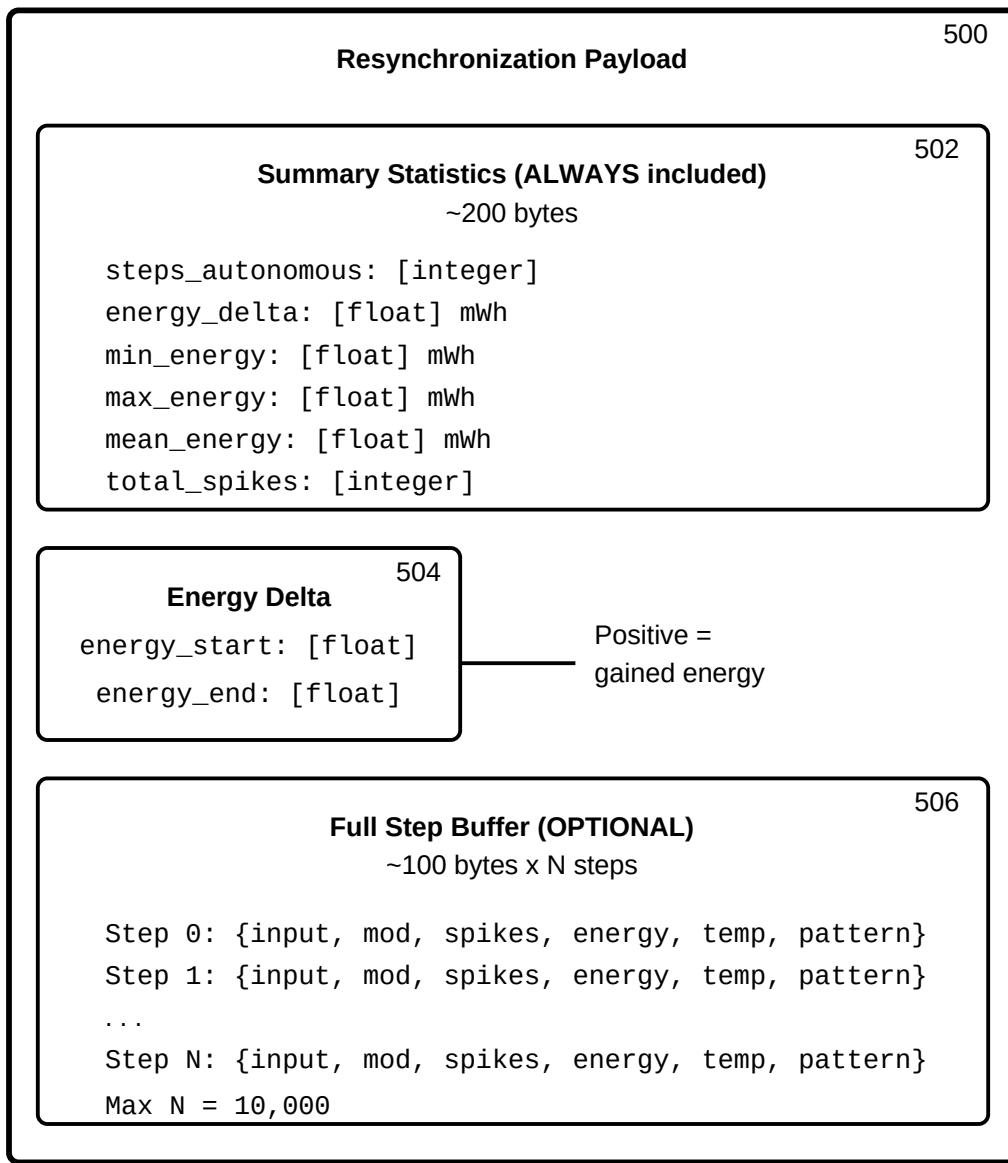
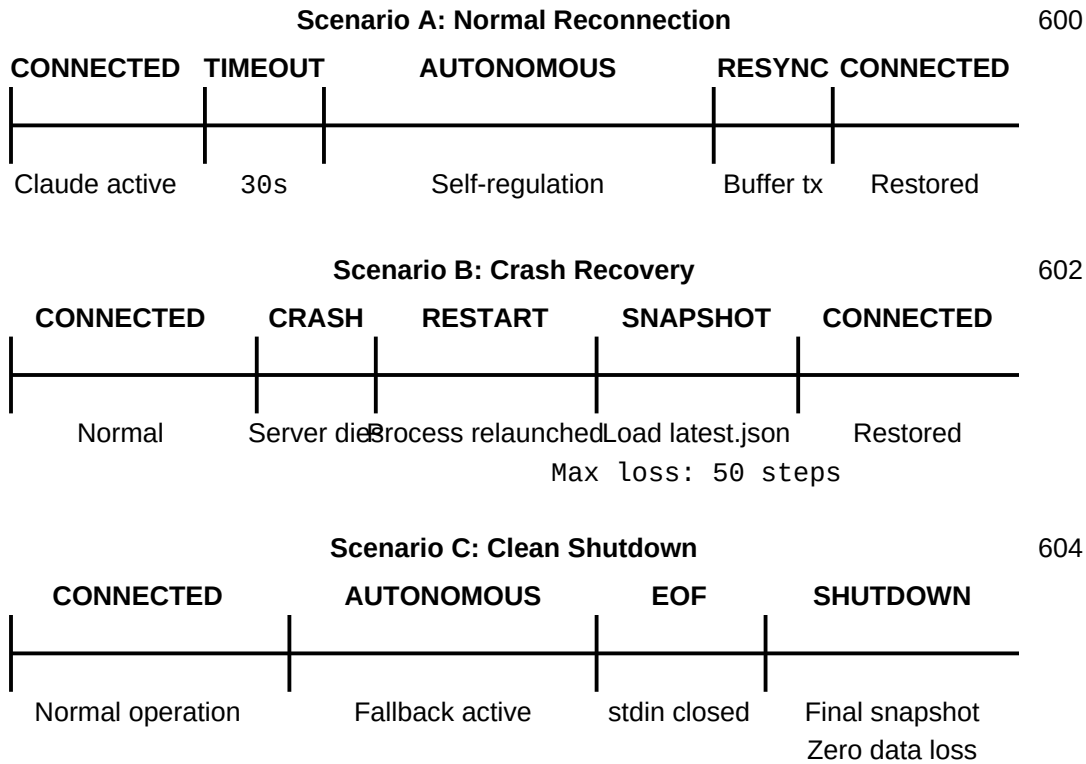


FIG. 5

RECOVERY SCENARIOS



Recovery Guarantees

606

Normal reconnection: Full continuity, resync payload transmitted

Crash recovery: Max 50 steps lost (snapshot every 50 steps)

Clean shutdown: Zero data loss (final snapshot on EOF)

Max autonomous duration: 10,000 steps (~50 hours)

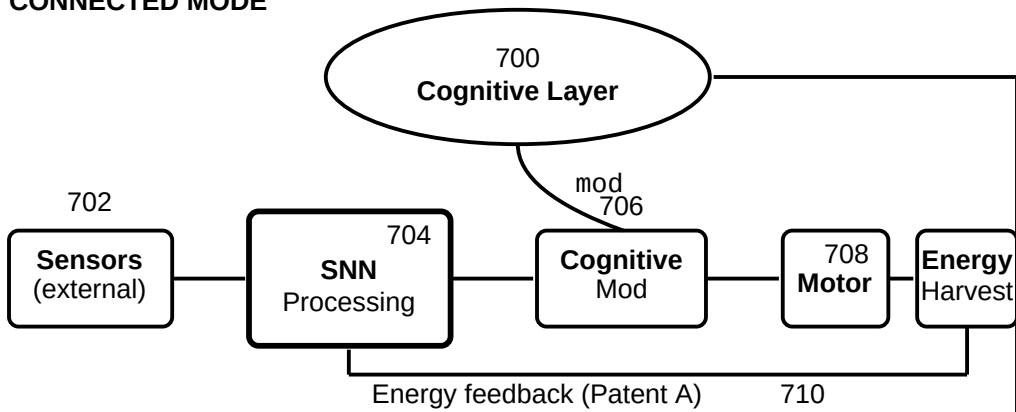
———— System actively processing

FIG. 6

END-TO-END SIGNAL FLOW

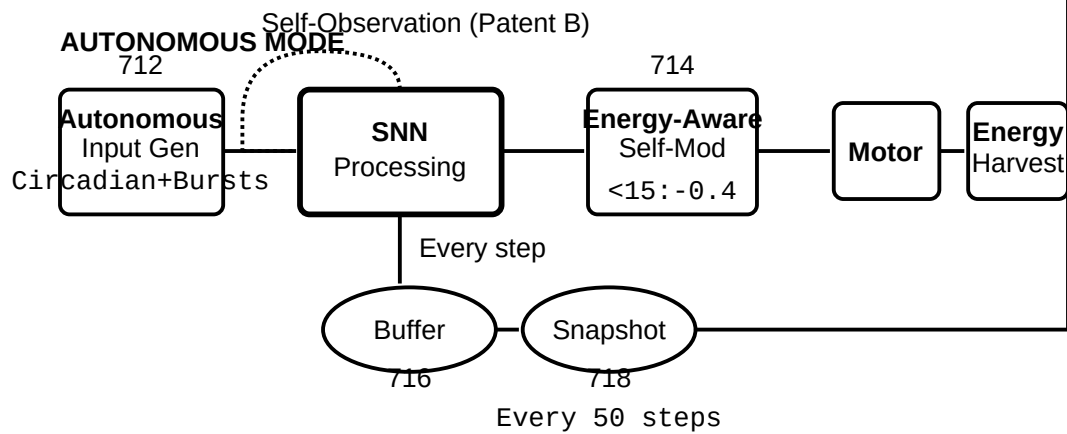
Connected vs. Autonomous Operation

CONNECTED MODE



DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - no interruption to SNN or Harvester

AUTONOMOUS MODE



Signal Pathways

- Forward signal path
- - - - - Energy/control feedback
- Self-Observation feedback (Patent B)

FIG. 7